WRITE UP

WEEK-2 CYBERSEC



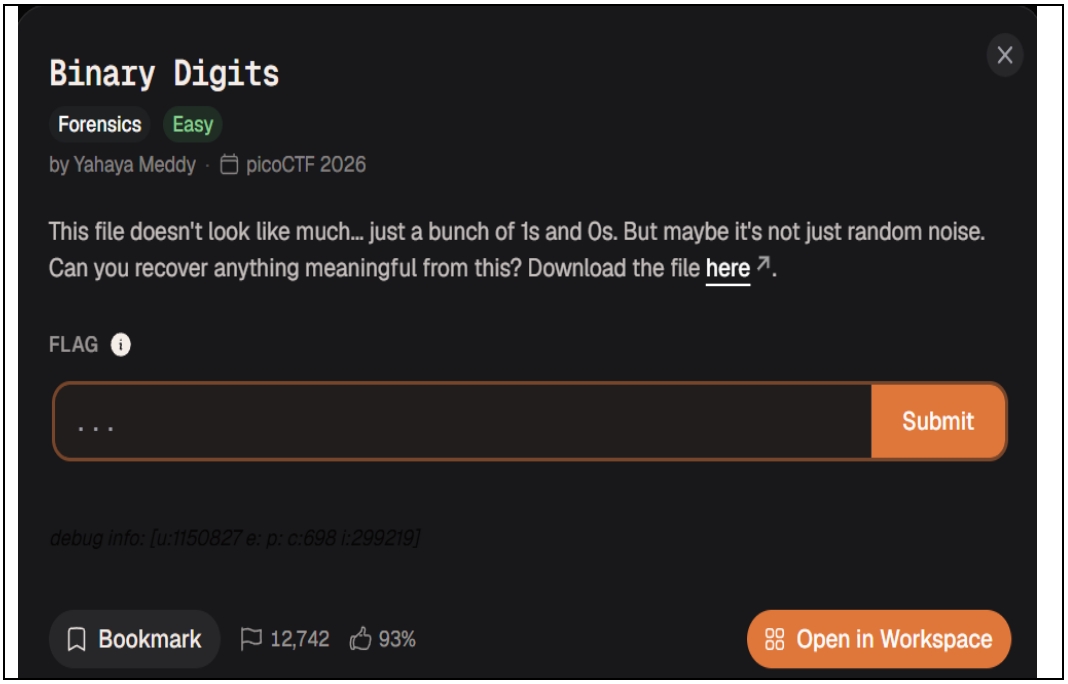
Nama : I Nyoman Devananda Satria Wibawa

Nim : 260530911110

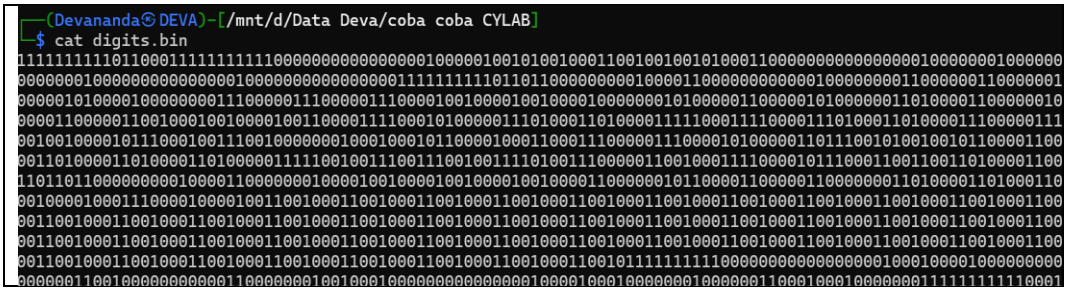
Divisi : Cyber Security

Forensic Challenge

- **Binary Digits**
  1. Download file yang diberikan pada menu challenge

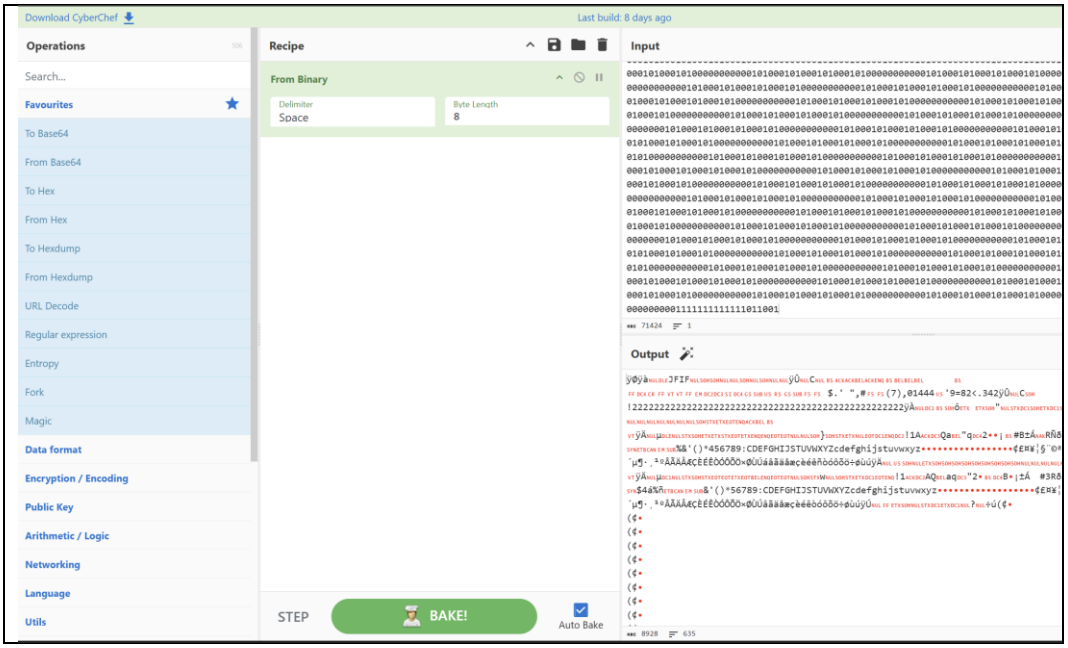


  2. Lalu coba buka file yang Bernama digits.bin dengan menggunakan command “cat” pada wsl



Maka file tersebut hanya berisikan angka binary saja

  3. Kita akan mengubah angka ini dengan cyberchef

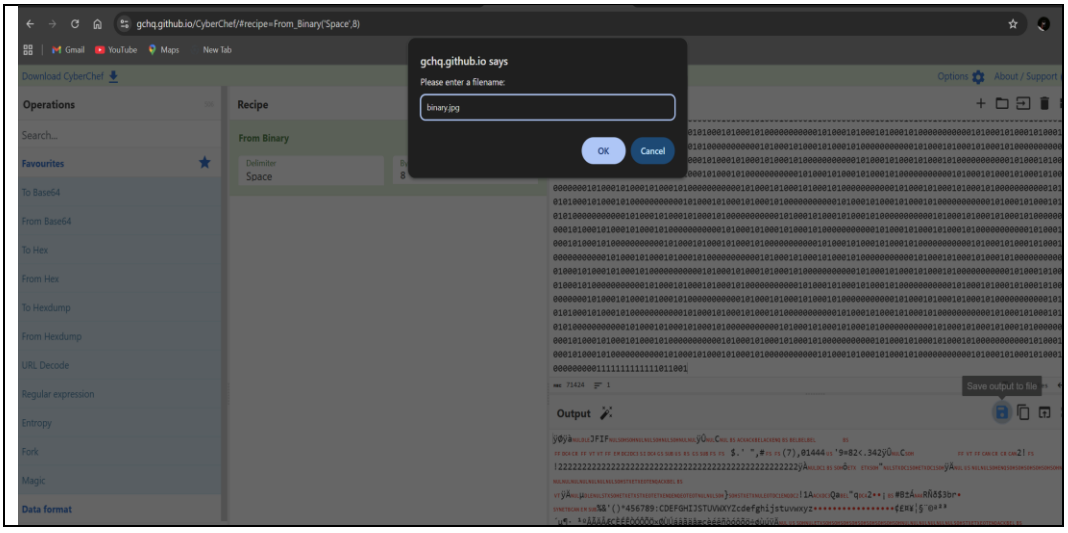


Copy angka binernya ke cyberchef

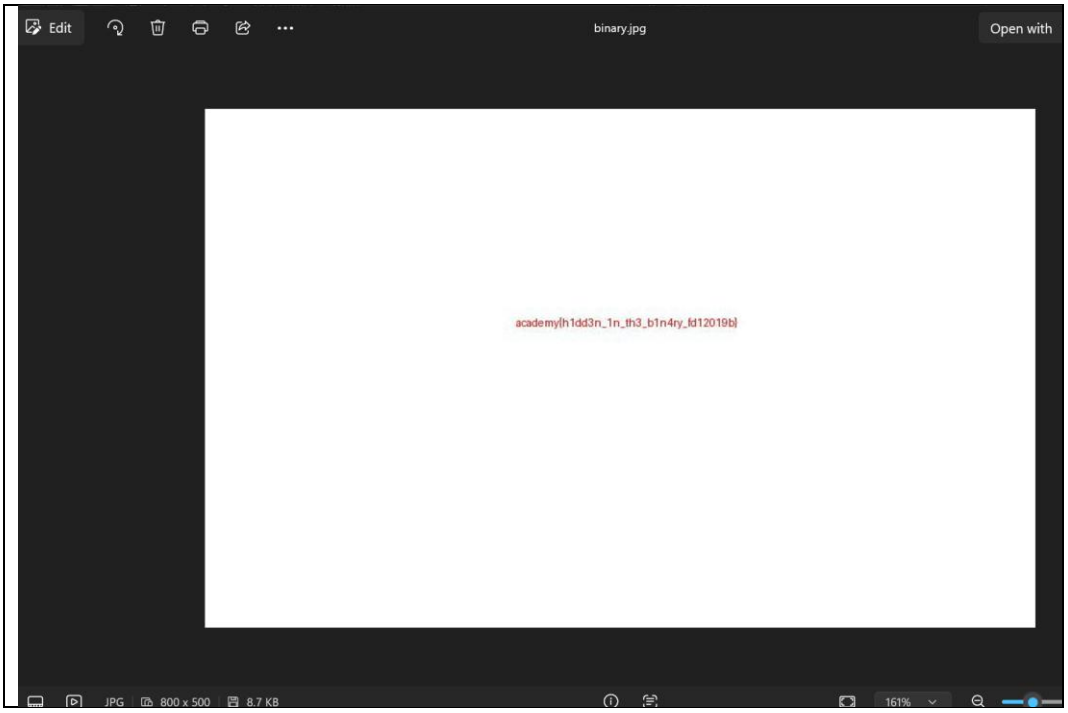
4. Pada bagian output terdapat JFIF yang artinya format file gambar pada linux



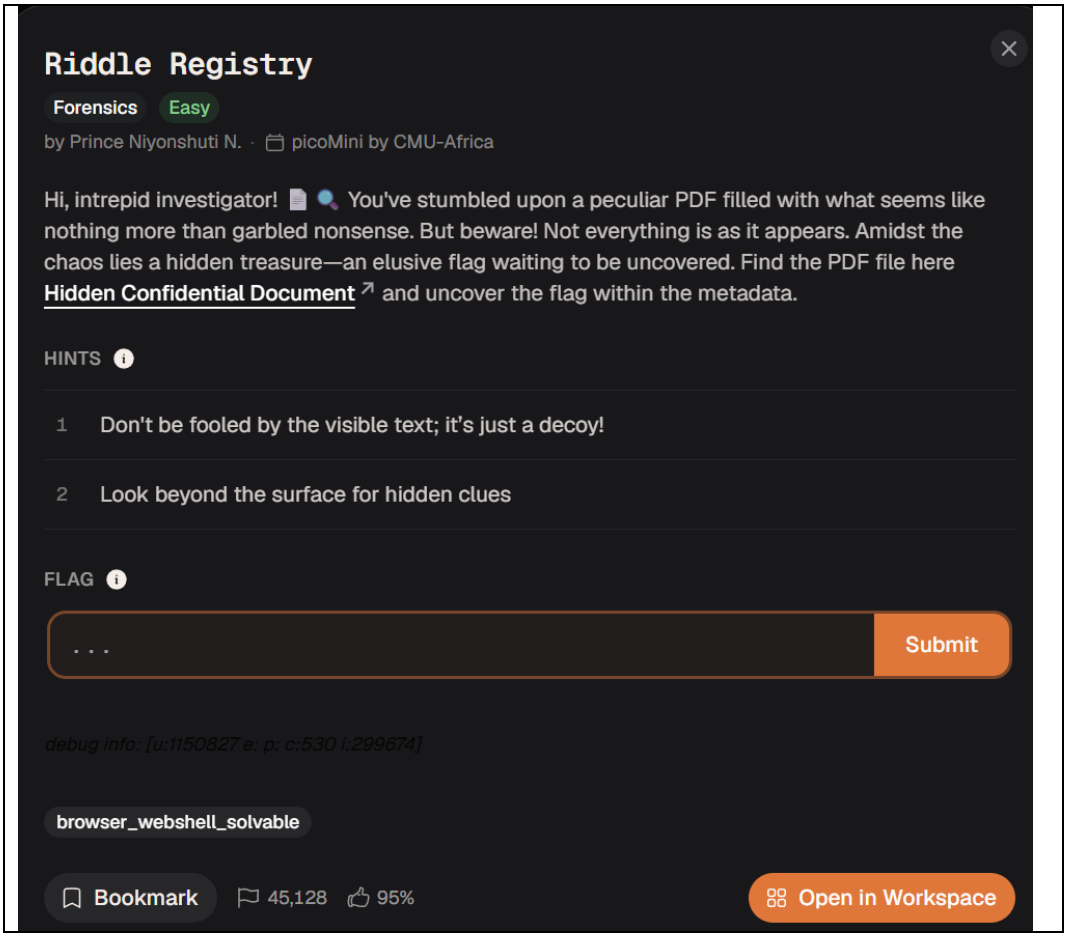
5. Lalu kita akan coba mendownload outputnya melalui cyberchef



6. Lalu buka gambar yang sudah didownload maka flag dapat ditemukan

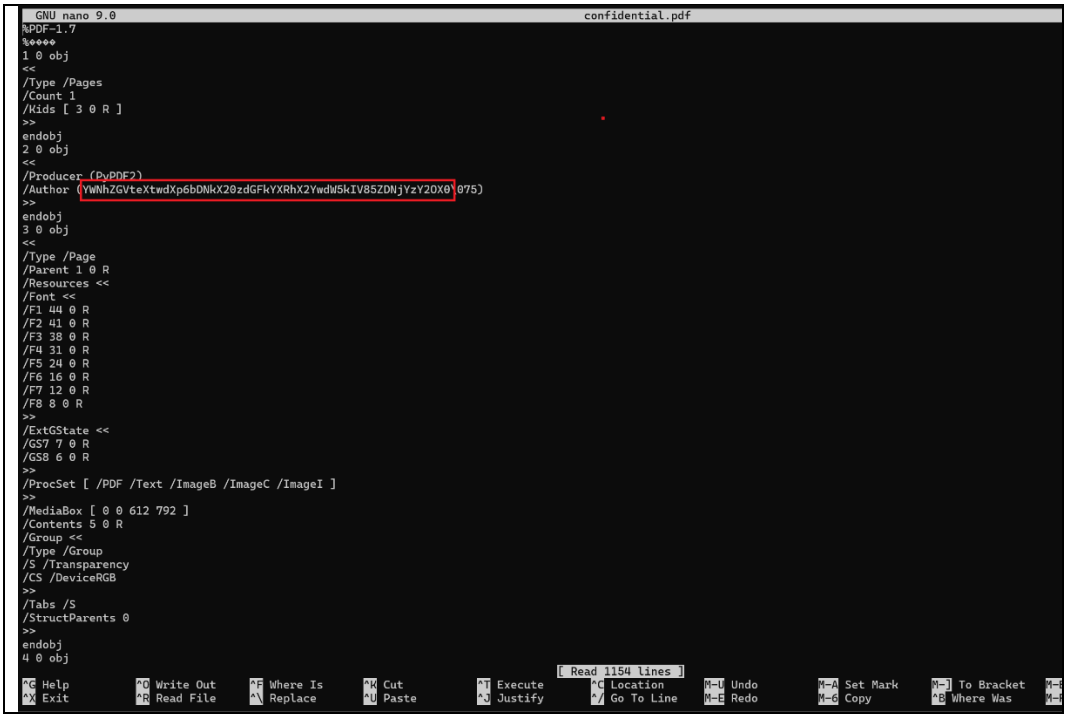


- **Riddle Registry**
  1. Buka menu challenge di cylab



2. Buka file confidential.pdf dengan command “nano” pada wsl

```
┌──(Devananda㉿DEVA)-[/mnt/d/Data Deva/coba coba CYLAB]
└─$ nano confidential.pdf
```



Hmmmm sepertinya ada teks yang mencurigakan pada bagian author

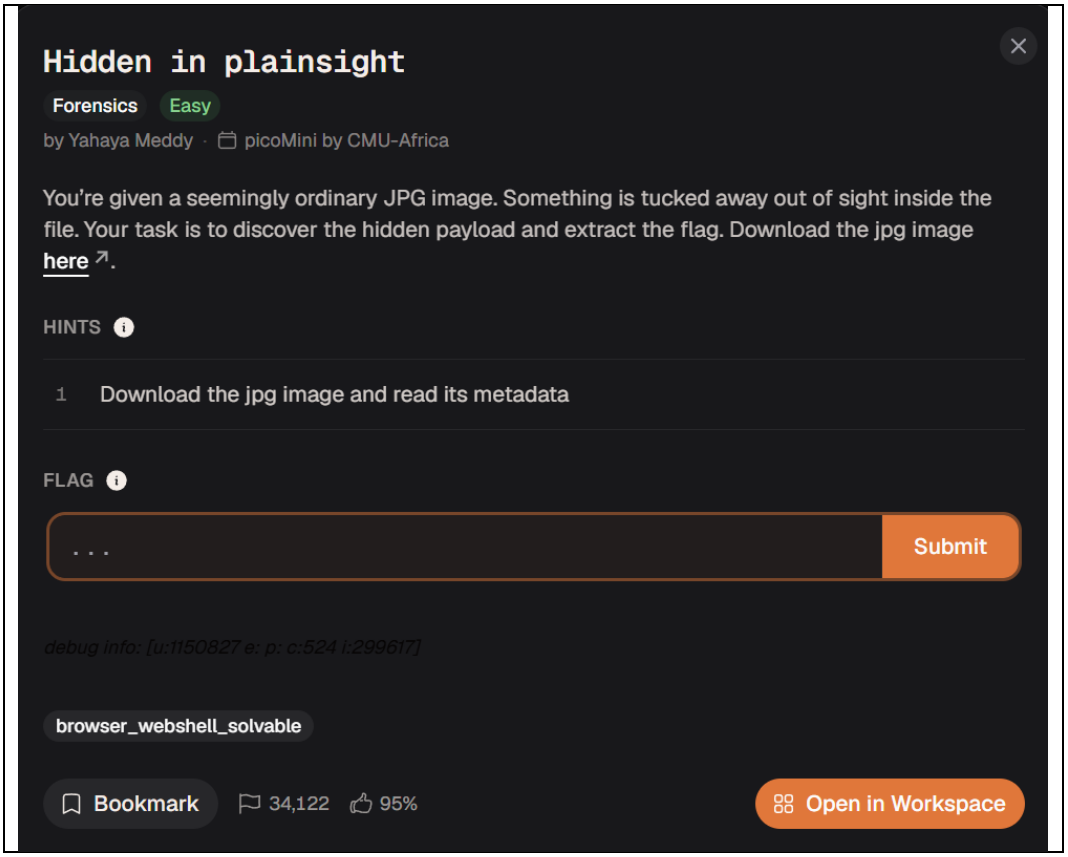
3. Coba decode teks tersebut menggunakan base64

```
┌──(Devananda㉿DEVA)-[/mnt/d/Data Deva/coba coba CYLAB]
└─$ echo "YWNhZGVteXtwdXp6bDNkX20zdGFkYXRhX2YwdW5kIV85ZDNjYzY2OX0" | base
academy{puzzl3d_m3tadata_f0und!_9d3cc669}
```

4. Maka flag akan didapatkan, jika sudah masukkan flag ke menu challenge

- **Hidden in plainsight**

1. Pada menu challenge terdapat sebuah hint untuk membaca meta data pada foto yang di download

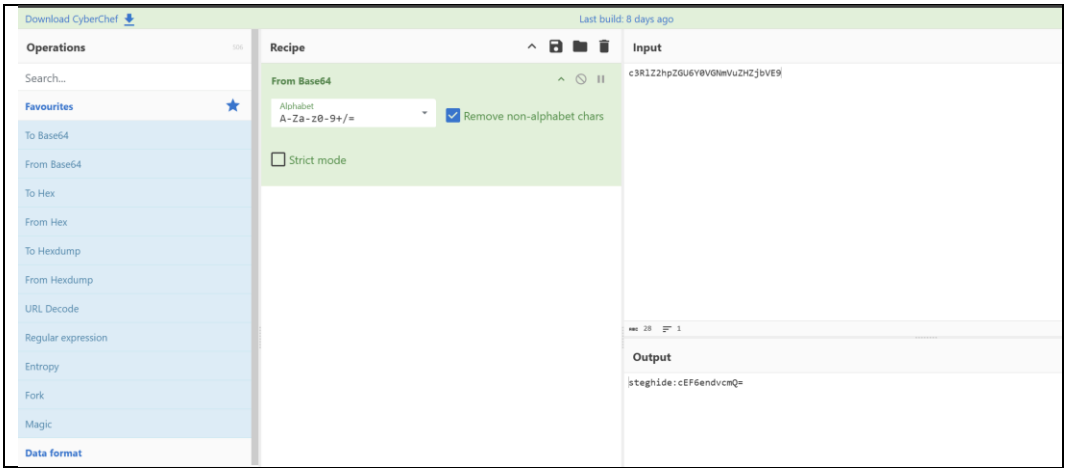


2. Kita gunakan exiftool untuk mencari clue selanjutnya

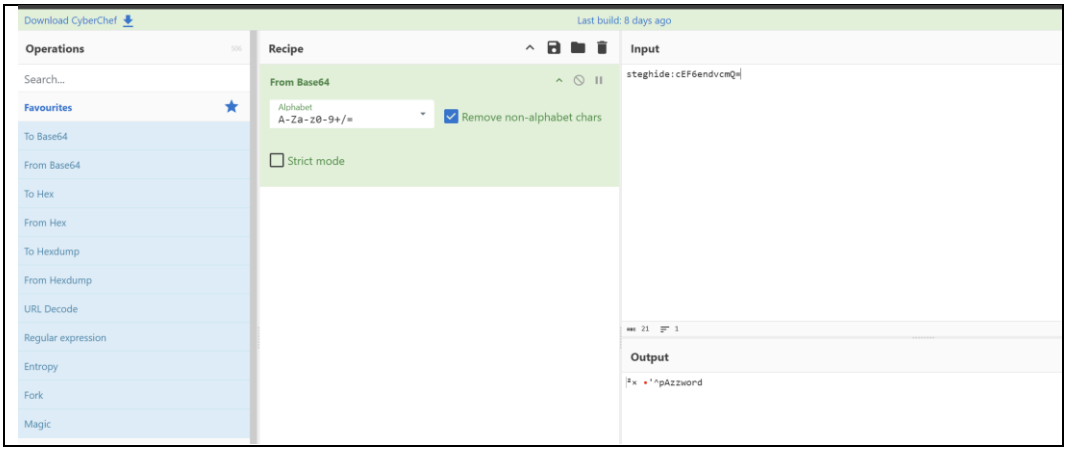
```
┌──(Devananda㉿DEVA)-[/mnt/d/Data Deva/coba coba CYLAB]
└─$ exiftool img.jpg
ExifTool Version Number         : 13.55
File Name                       : img.jpg
Directory                       : .
File Size                       : 74 kB
File Modification Date/Time     : 2026:09:30 15:11:46+08:00
File Access Date/Time           : 2026:10:11 15:57:28+08:00
File Inode Change Date/Time     : 2026:09:30 15:11:54+08:00
File Permissions                : -rwxrwxrwx
File Type                       : JPEG
File Type Extension             : jpg
MIME Type                       : image/jpeg
JFIF Version                    : 1.01
Resolution Unit                 : None
X Resolution                    : 1
Y Resolution                    : 1
Comment                         : c3RlZ2hpZGU6Y0VGNmVuZHZjbVE9
Image Width                     : 640
Image Height                    : 640
Encoding Process                : Baseline DCT, Huffman coding
Bits Per Sample                 : 8
Color Components                : 3
Y Cb Cr Sub Sampling            : YCbCr4:2:0 (2 2)
Image Size                      : 640x640
Megapixels                      : 0.410

┌──(Devananda㉿DEVA)-[/mnt/d/Data Deva/coba coba CYLAB]
└─$
```

3. Gunakan cyberchef untuk mendecode teks pada bagian comment dari base64



4. Copy output pertama lalu decode lagi dan kita akan mendapatkan sebuah password



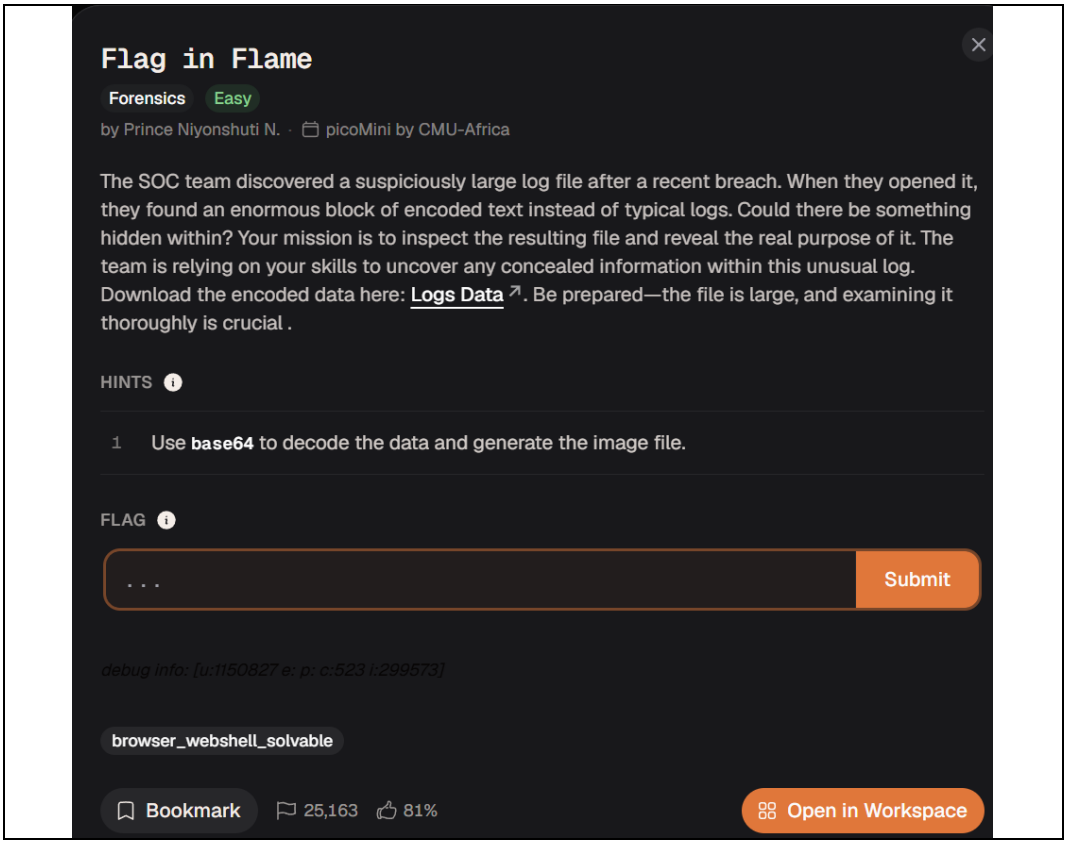
5. Karena kita sudah mendapatkan passwordnya, maka kita akan coba untuk mengekstrak img.jpg menggunakan command steghide dan masukkan password

```
┌──(Devananda㉿DEVA)-[/mnt/d/Data Deva/coba coba CYLAB]
└─$ steghide extract -sf img.jpg
Enter passphrase:
wrote extracted data to "flag.txt".
```

6. Gunakan “cat” untuk membuka file flag.txt dan flag pun berhasil didapatkan

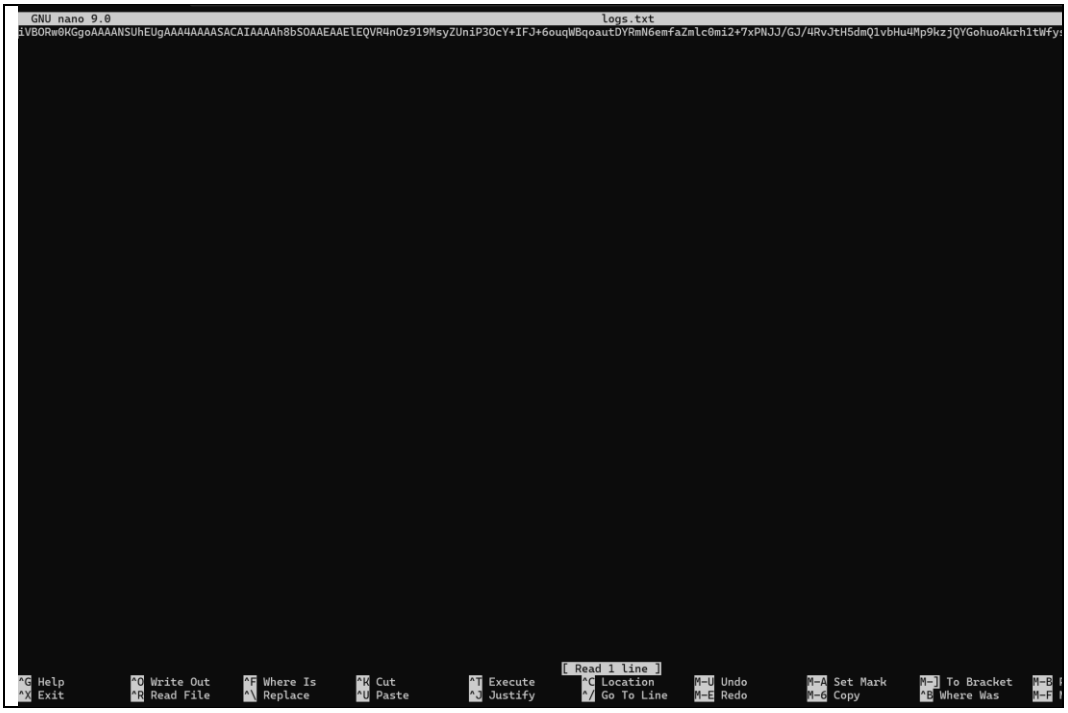
```
┌──(Devananda㉿DEVA)-[/mnt/d/Data Deva/coba coba CYLAB]
└─$ cat flag.txt
academy{h1dd3n_1n_1m4g3_7ba83d11}
```

- **Flag in Flame**
  1. Download file pada menu challenge



2. Buka file menggunakan nano maka akan terdapat sebuah text




GNU nano 9.0 logs.txt
iVBORw0KGgoAAAANSUhEUgAAA4AAAASACAIAAAAh8bSOAAEAAElEQVR4nOz919MsyZUniP2eiP2ZuAjQ+6RQ+gEwiP2ZuAjQ+6RQ+gEwiP2ZuAjQ+6RQ+gEw

3. Lalu kita decode teksnya menggunakan base64 dan akan muncul seperti ini



4. Kita akan coba decode file logs.txt dan juga merubahnya menjadi format jpg

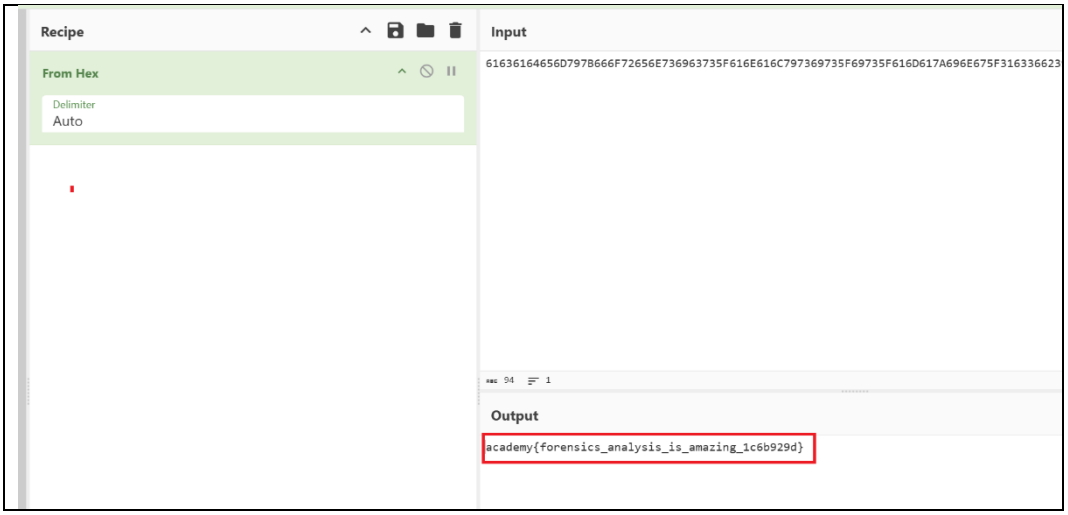


5. Kita coba buka filenya melalui folder



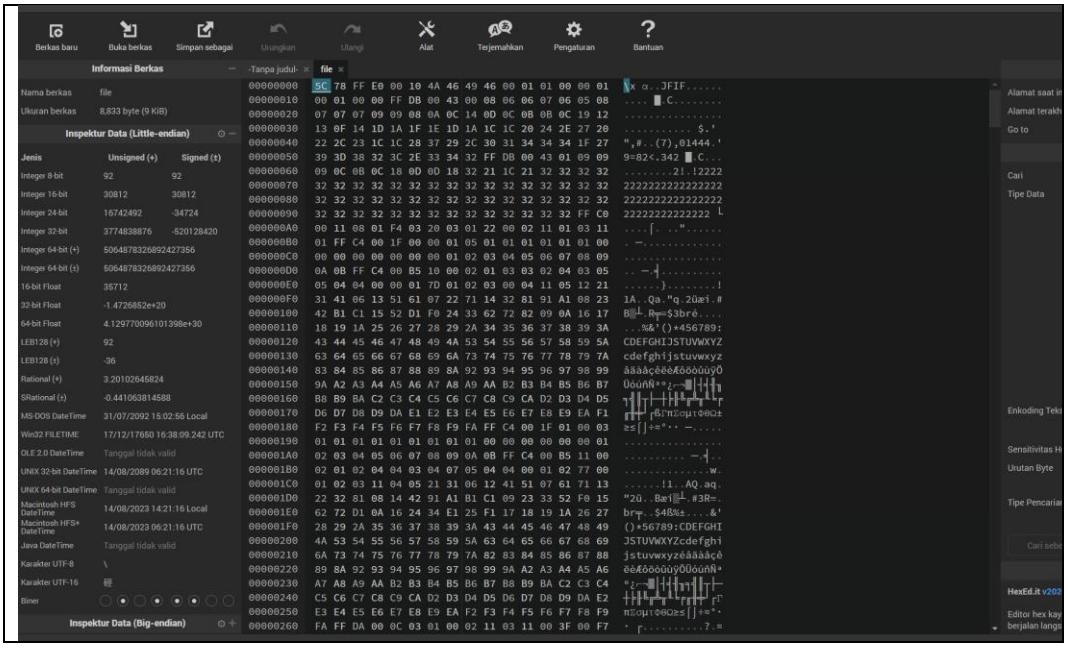
Maka akan muncul foto heker tampan dan didalamnya terdapat sebuah teks, hmmm

6. Kita coba decode teksnya menggunakan cyberchef dari hex



7. Flag sudah didapatkan

- **Corrupted File**
  1. Download file pada menu challenge
  2. Selanjutnya saya menggunakan web HexEd.it untuk membuka file dengan hex sesuai pada hint



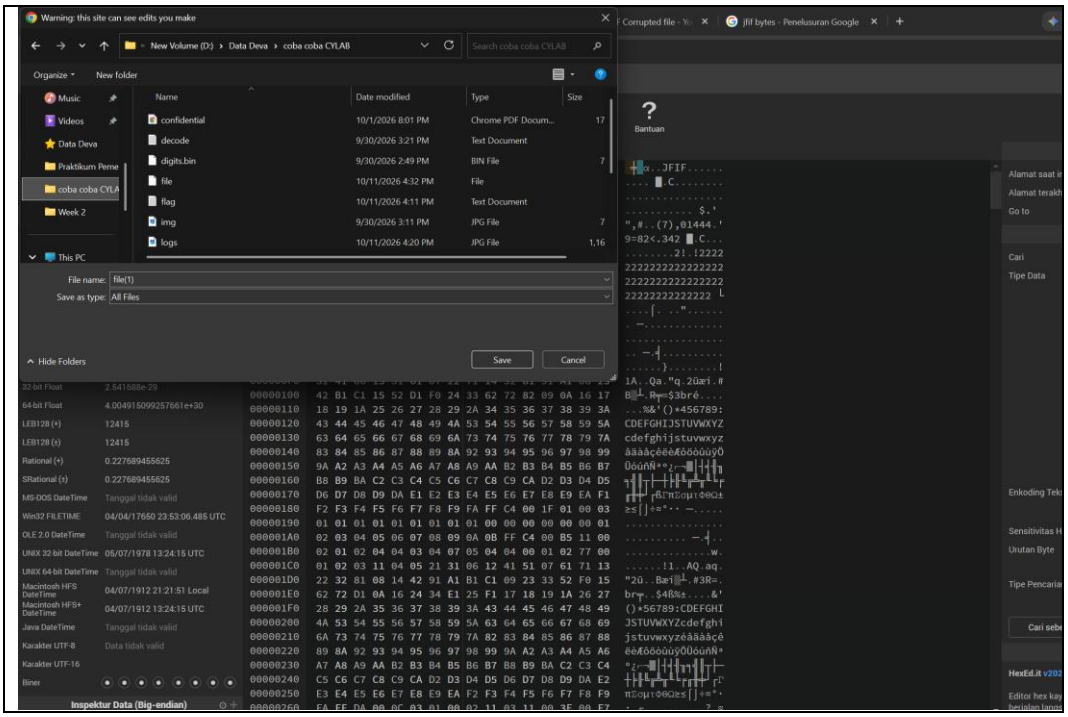
3. Kita lihat pada bagian jfif atau header. Jfif memiliki magic bytes yakni FF D8 FF E0 dan pada bagian header bytes tersebut tidak sesuai, maka kita akan Ganti bytesnya



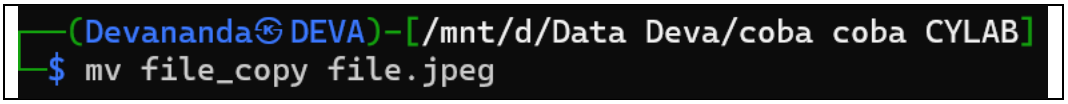
4. Ubah bytesnya



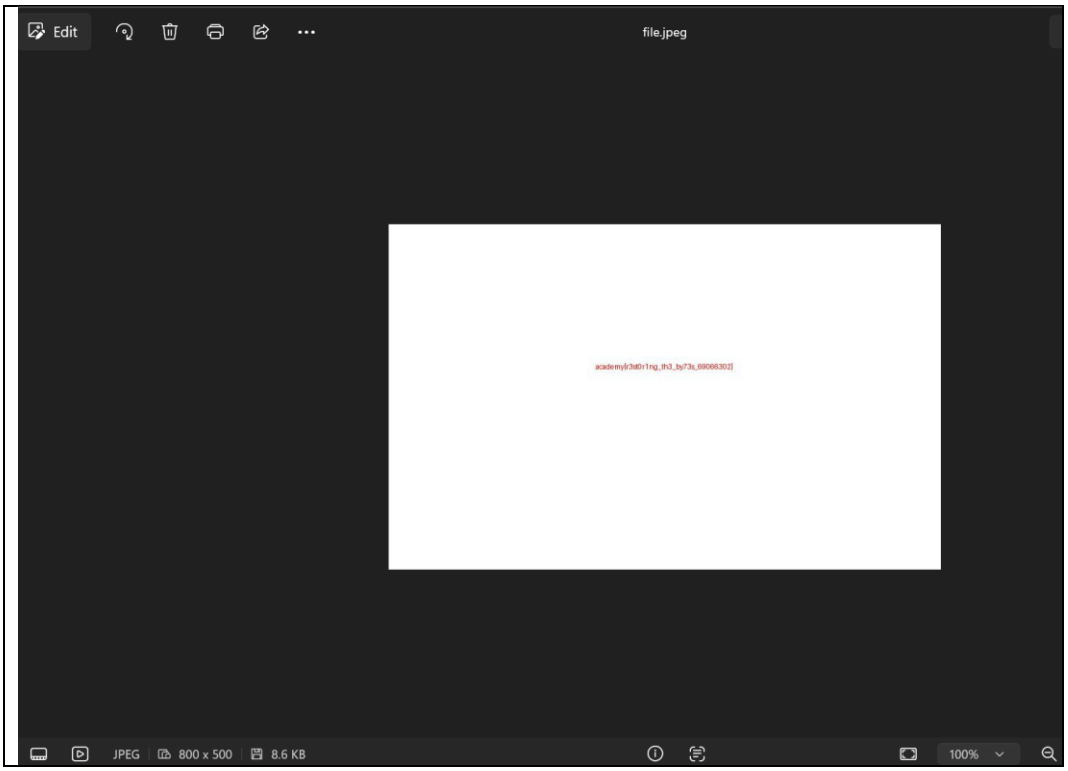
5. Export file yang sudah diubah bytesnya



6. Lalu kita ubah format filenya menjadi JPEG



7. Buka file jpeg nya dan flag akan ditemukan



Kategori Cryptography

- **StegoRsa**
  1. Downloag file image dan flag.enc yang disediakan
  2. Kita coba buka file image.jpg menggunakan exiftool

```
┌──(Devananda㉿DEVA)-[/mnt/d/Data Deva/coba coba CYLAB]
└─$ exiftool image.jpg
ExifTool Version Number         : 13.55
File Name                       : image.jpg
Directory                       : .
File Size                       : 21 kB
File Modification Date/Time     : 2026:10:12 11:40:23+08:00
File Access Date/Time           : 2026:10:12 11:41:44+08:00
File Inode Change Date/Time     : 2026:10:12 11:41:07+08:00
File Permissions                : -rwxrwxrwx
File Type                       : JPEG
File Type Extension             : jpg
MIME Type                       : image/jpeg
JFIF Version                    : 1.01
Resolution Unit                 : None
X Resolution                    : 1
Y Resolution                    : 1
Comment                         : 2d2d2d2d2d424547494e2050524956415445204b45592d2d2d2d2d0a4d4949457641494241444414e42676b71686b694739773044
b597767675369416745414164f4942415143734d6179734e586530784b45720a3343786a75324d355a384c6a485876684c767674717266c6c376c79794a4576395978356135
647472636f4f6533644162377646368376d0a4b566f4a4b614b4d447259695737725753 6b786e732f392b616c52744b6d4a4f767875474f47562b6e453765724d4d7669497
76253790a6f5a5546344c6b58744e67656738436364765146453749755549375 4325a6b5759552b4c6f2b585279315830336f6c6338614135504e4b6b427479667234767 9
2b313449394b51584b6c744c306c623264366651554658552b614f6c596a64514e53654f31413072374857151 4d54717a646d6f5a78390a49766a6364736236385 14e706
2516d7359336a7854435574e306a4d7544707331674 36b45385365755a5853454874775a4f5337474d620a306b3449594473394167 4d42414145436767454142566707259
764b752f646d64462f614d2b664a4b7a2b6162497 4b556d504b4f2f0a473462766c625750624 24a467a357071514 54c79394b314f75476c394b4465686e6b33524d79504
84d32396c7572654c563959383 4380a6d5655737076754450537973572b457a6b6366785263694 96b70663971316d6c707233523437314e2f655769703656626e524b5154
660a5455645175777255432b5671706b4273774a3231565638594a706174377058 4d4e5a6169556b4d43684a766a6d4d4c676c4248506f742b614a634f62724570340a4d7
548786f794f592b6668384c626932585138487965546d476e5070444265487635387 56e75784d3550676a4e6854556b6e677573350a4e4879623168325569363331635a76
766b557647306a4a44565344f2f446c74774b4267514471763862456f7855764b6c5751547154 2b0a484371776d32782f4a3043707032517269626d73784b47626f3042436
f613173776c64506453526655585 07a4c766c6249497134676d0a665638755879 4e50595350692b5763457a706d726472472b5a7567566f364e3851334b78327973686c2b
574635594e4d6877512b48370a4b4d4176713532444e524b54596d36744e6c56774e3465534e774b4267514337794 4554f7a306c4251614475497339334733566386 4a543
6305133744163335046 6d446e3758506a4a48585753766f4450682b3155435233 2b777973706a794c6d4e7867644a746945346c324761695963367 97a33486e0a6b384567
3546617a47755958745 3416e6d7866766666692f77735a496e306f2b2b634e587644776e6d456a63304 734d6e733141794 70a4e532b34397053304b774b4267466b49553
831436273562b53546a68724c73474b6e38663378526c4b5959524b2b582b6a5164505352 0a5a50412b674c59424d58776a57496f44726d6b6174783758443662714a3845
685a4873534c4a6d4d3748506 4b524a5154514113 6c6c0a75324a705658707744f442b6730673674727852 2b3734 6c55753543597 26d53726c2f6838686b6a31494135324
352416f4741513939780a6a68444f59592f66497a506832534e4c7139626c6773646a444 4e5a632f6c7378785243464b796e6445503542 4d4f75574b3068483 6e41527759
394a4977763157386 4b4167456478 6c6f7633697178576246765 0a693733962357a334e6c4a4f742b53596b64512f4a6f4b4a524d65434e305a4a43730a6254394a76657
a7565534732527a462b37744e2f5578637146747a6c45436759425749363 6b654b5850623636554465034363546470a7466756159435030526b5a7764536 2b737a644b
3053342f69777746d4b42676f384d50363059572b797777302f73356a534c344450680a4468343167 4b4337373847583737 4a4f625638534f486b44557a6255456d4e72526
042536e663555515338496c6d3873792f38376c440a41624f325962355847705634464167 4b4e6d75737a773d3d0a2d2d2d2d2d454e4420505249564154 4520b45592d2d
Image Width                     : 512
Image Height                    : 512
Encoding Process                : Baseline DCT, Huffman coding
Bits Per Sample                 : 8
```

  3. Selanjutnya decode from hex menggunakan web cyberchef

```
-----BEGIN PRIVATE KEY-----
MIIEvAIBADANBgkqhkiG9w0BAQEFAASCBKYwggSiAgEAAoIBAQCsMaysNXe0xKEr
3Cxju2M5Z8LjHXvhLvvtqrll7lyyJEv9Yx5a5yFZbHrxY4OdtrcoOe3dAb7vF87m
KVoJKaKMDrYiW7rWSkxns/9+alRtKmJOvxuGOGV+nE7erMMviIwx0tqxYk6KgbSy
oZUF4LkXtNgeg8CcdvQFE7IuUI7T2ZkWYU+Lo+XRy1X03olc8aA5PNKkBtyfr4vy
dwOwU7odi+14I9KQXKltL0lb2d6fQUFXU+aGlYjdQNSeO1A0r7HEqQMTqzdmoZx9
Ivjcdsb68QNpa//VX1iCZfbQmsY3jxTCWN0jMuDps1gCkE8SeuZXSEHtwZOS7GMb
0k4IYDs9AgMBAAECggEABVfprYX5Oxa3XUBdvKu/dmdF/aM+fJKz+abItKUmPKO/
G4bvlbWPbBJFz5pqQELy9K1OuGl9KDehnk3RMyPKHTUfQy18IxM29lureLV9Y848
mVUspvuDPSysW+EzkcfxRciIkpf9q1mlpr3R471N/eWip6VbnRKQTvbSzcVTl7cf
TUdQuwrUC+VqpkBswJ21VV8YJpat7pXMNZaiUkMChJvjmMLglBHPot+aJcObrEp4
MsUfBWHE+ch5HxoyOY+fh8Lbi2XQ8HyeTmGnPpDBeHv58unuxM5PgjNhTUkngus5
NHyb1h2Ui631cZv4+JoS0o9wIvkUvG0jJDVSO/DltwKBgQDqv8bEoxUvKlWQTqT+
HCqwm2x/J0Cpp2QribmsxKGbo0BCae+81Nc7RvOa1swldPdSRfUXPzLvlbIIq4gm
fV8uXyNPYSPi+WcEzpmrdrG+ZugVo6N8Q3Kx2yshl+GDLuZewEkwWF5YNMhwQ+H7
KMAvq52DNRKTYm6tNlVwN4eSNwKBgQC7yDUOz0lBQaDuIs93G3V8cJT75ynN9N3m
f0Q3tAc3PFmDn7XPjJHXWSvoDPh+1UCR3+wyspjyLmNxgdJtiE4l2GaiYc6yz3Hn
k8EgWw6uRIYZQJ5FazGuYXtSAnmxfvffi/wsZIn0o++cNXvDwnmEjc0FsMns1AyG
NS+49pS0KwKBgFkIU0LQ93/j9n+81CbsV+STjhrLsGKn8f3xRlKYYRK+X+jQdPSR
ZPA+gLYBMXwjWIoDrmkatx7XD6bqJ8ExW6AixSmeFhZHsSLJmM7HPbKRJQTQA3ll
u2JpVXpwOD+g0g6trxR+74lUu5CYrmSrl/h8hkj1IA52DrPzH4RoCysRAoGAQ99x
jhDOYY/fIzPh2SNLq9blgsdjDNZc/lsxxRCFKyndEP5BMOuWK0hH3nARwYZC+Hb/
YLO9JIwv1W8fKAgEdxlov3iqxWbFvPZi79b5z3NlJOt+SYkdQ/JoKJRMeCN0ZJCs
bT9JvesjSErsmS1JzueSG2RzF+7tN/UxcqFtzlECgYBWI63keKXPb66UDeP465FG
tfuaYCP0RkZwdSe+szdK9nXz/PcKBf0S4/iwtmKBgo8MP60YW+yww0/s5jSL4DPh
Dh41oKC778GX77JObV8SOHkDUzbUEmNrRiKoJ7tgLb40BSnf5UQS8Ilm8sy/87lD
AbO2Yb5XGpV4FAgKNmuszw==
-----END PRIVATE KEY-----
```

4. Lalu decrypt dengan opeSSL

```
┌──(Devananda㉿DEVA)-[/mnt/d/Data Deva/coba coba CYLAB]
└─$ openssl pkeyutl -decrypt -inkey "private.pem" -in "flag.enc" -out "fl

┌──(Devananda㉿DEVA)-[/mnt/d/Data Deva/coba coba CYLAB]
└─$ ls | grep pri
private.pem

┌──(Devananda㉿DEVA)-[/mnt/d/Data Deva/coba coba CYLAB]
└─$ ls | grep flag
flag.enc
flag.txt

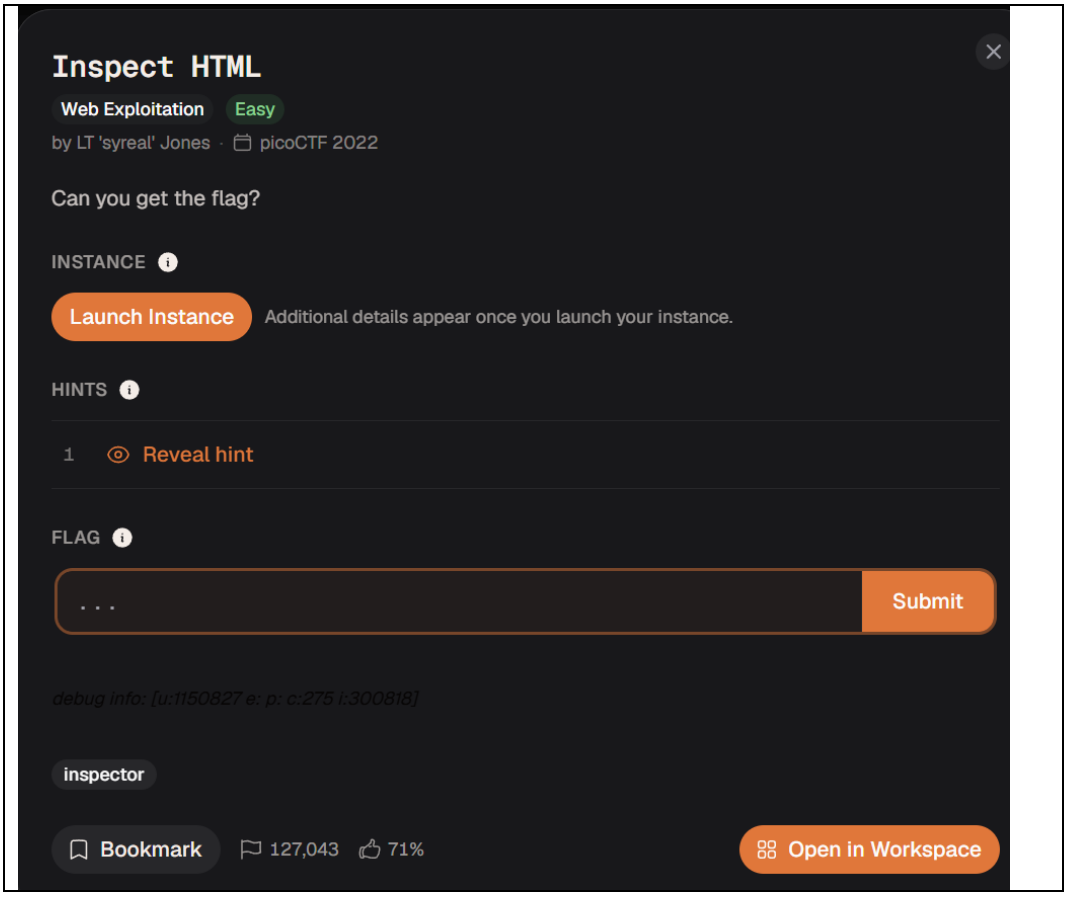
┌──(Devananda㉿DEVA)-[/mnt/d/Data Deva/coba coba CYLAB]
└─$ cat flag.txt
academy{rs4_k3y_1n_1mg_d97b7bfb}
```

5. Flag sudah didapatkan dan bisa dimasukkan ke cylab

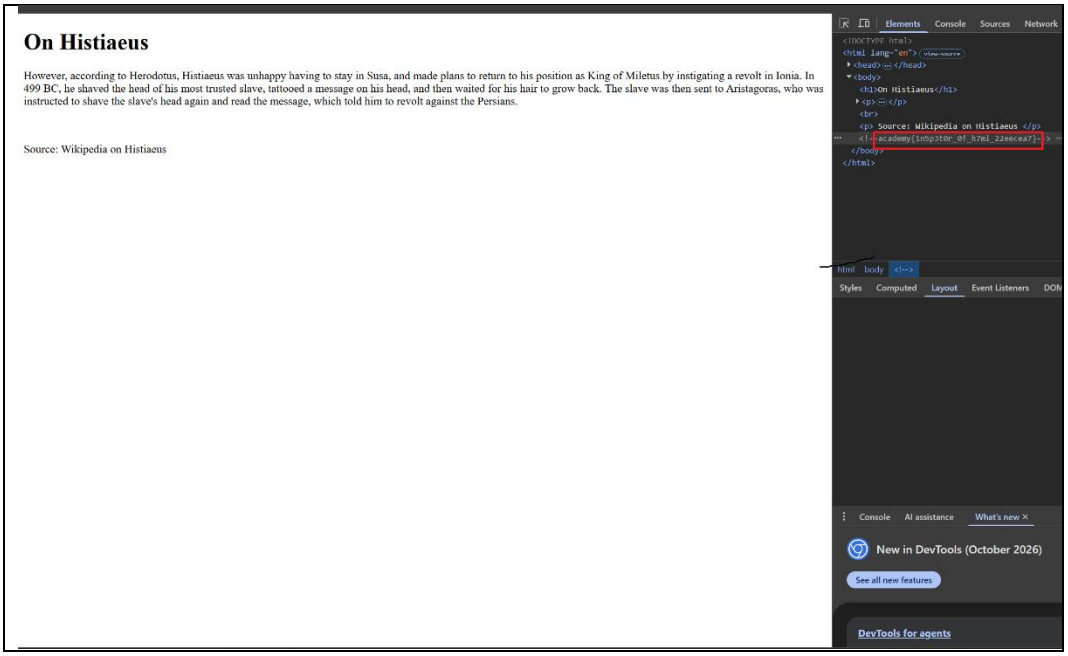
Kategori Web Exploitation

- **Inspect HTML**

1. Jalankan instance pada menu challenge



2. Buka website yang disediakan lalu kita inspect menggunakan shortcut F12



3. Flag sudah didapatkan dan bisa dimasukkan ke cylab